

PATRICK BEAL

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Data scientist and economist who builds production systems. Works across optimization, machine learning, Bayesian methods, and econometrics, with the past year spent as a founding engineer taking products from a first idea to working software people rely on. Pairs careful, rigorous analysis with clear explanation, for research teams and executives alike.

EDUCATION

Applied and Computational Mathematics (ACME) and Economics, BS

Apr 2026

Brigham Young University

- GPA: 3.92 | Wheatley Scholar (2024 Cohort) | Dean's (4x)
- Coursework: Optimization, Algorithms, Machine Learning, PDEs/ODEs & Optimal Control, Econometrics, Software Development

EXPERIENCE

Founding Engineer — TechForce Advisors (Remote)

August 2025 – Present

- Built and shipped a production agentic research system that turns a visitor's email into a personalized company report, with full CI/CD, a cost-monitoring dashboard, and a verifier pass that strips unsupported claims before anything ships (about 78 seconds and \$0.10 in LLM cost per report) (Next.js, AWS Bedrock AgentCore, Claude)
- Built and shipped a native iOS app (SwiftUI) that transcribes a consultant's voice note (AWS Transcribe), extracts the company, contact, and a summary of the discussion (Bedrock Claude), and writes them directly into the CRM, authenticated via SAML SSO through a custom-scheme deep-link callback (Swift, AWS Transcribe, Bedrock)
- Extended that research system into an automated outbound pipeline that sources prospects, pre-generates a report for each in parallel, and emails them with automated bounce detection and suppression to protect deliverability, tracking real engagement (dwell time, link clicks) back into the CRM (Apollo, AWS, Gmail API, GoHighLevel)
- Designed and built a shared AWS test-infrastructure platform for three production apps (Terraform-managed VPC, Aurora Serverless v2, host-based ALB routing), and traced a production outage to a broken health check, fixing it by swapping curl for wget (Terraform, AWS ECS Fargate, Aurora Serverless v2)
- Currently building an AI pipeline that extracts financial data from real estate broker documents (offering memos, rent rolls, financials) into acquisition underwriting models, using document parsing and grounded LLM extraction (Docling, AWS Bedrock, Python)

GTM Strategy & AI Intern — Winning by Design, Provo, UT

May 2025 – September 2025

- Engineered a full-stack application that automated the Go-To-Market Diagnostic, standardizing CRM, finance, and customer success data into 20+ analyses (Python, Flask, SQL, Google APIs)
- Collaborated with a 3-person team and presented results to executives, reducing diagnostic turnaround from 6–8 weeks to 5–15 minutes
- Designed and deployed an LLM-based “Deep Dive” pipeline to analyze client call transcripts, scaling review capacity from about 20 calls over several weeks to as many calls as the model API could process in under 30 minutes
- Automated slide-deck generation and delivered insights through a Slack interface, enabling consultants to advise clients with consistent, data-driven outputs

Research Assistant — Office of the President, BYU

August 2024 – December 2025

- Research and author reports for President Reese on student development and related topics, synthesizing scholarship and literature into actionable insights to inform university strategy
- Collaborate with 6 research assistants and present findings regularly to the president's office, strengthening research through peer review and executive feedback

Research Assistant — Department of Mathematics, BYU

December 2023 – October 2024

- Design and evaluate algorithms for biomarker detection in photoplethysmogram (PPG) signals, including Bayesian modeling, peak detection, signal quality metrics, mixture-of-experts ensembles, and diastolic notch detection, with implementations in Python and integration into the lab's proprietary package
- Conduct experiments and comparative analyses in Jupyter notebooks, apply Bayesian methods to improve signal reliability, and document research workflows to support team reproducibility and onboarding

Opera Accompanist — School of Music, BYU

May 2025 – April 2026

- Prepare and perform operatic repertoire at a professional level, coordinating closely with conductors and vocalists to achieve seamless ensemble performance in rehearsals and productions
- Support vocalists by balancing dynamics and timing, highlighting transferable strengths in attentive listening, precision, and team collaboration within high-stakes environments

Ballet Accompanist — *Department of Dance, BYU*

August 2024 – April 2026

- Perform live accompaniment for 2-6 weekly ballet classes, including BYU's two premier company groups, by preparing, improvising, and sight-reading music across 8+ styles to support technique and choreography
- Adapt in real time to professor direction and dancer pacing, demonstrating strong situational awareness, rapid problem solving, and collaborative execution under pressure

Additional Experience & Leadership

- **Mathematics Grader (Math 290)**, BYU (*Aug – Dec 2023*): Graded and guided 20 students on proofs in logic and discrete math
- **Volunteer Representative**, The Church of Jesus Christ of Latter-day Saints, New York Syracuse Mission (*Aug 2021 – Aug 2023*): Automated monthly fuel-log reporting for the mission's roughly 100 vehicles (Google Apps Script)
- **Research Assistant**, BYU Whitehead Lab (*Apr – Aug 2021*): Documented Bayesian statistical sampling tests for geologic computing software (Python)
- **Brass Salesperson / Software Developer**, Summerhays Music Center (*2018 – 2021*): Automated payment entry with PyWinAuto, cutting entry time by 85%
- **Music Leadership**: Principal/Lead Trombonist, BYU Symphony Orchestra, Lyceum Philharmonic, and Jazz Ensemble — led section rehearsals, mentored peers, and ensured high-level performance across orchestral and jazz repertoires

PROJECTS

Football to Admissions – 1st Place, BYU Statistics Department Case Study Competition (2025)

- Developed pipelines (Python, pandas, scikit-learn, XGBoost) linking NCAA football records with College Scorecard admissions data to test the “Flutie effect”
- Implemented regression models, clustering, and engineered features (lagged win rates, tournament runs) to evaluate correlations between athletic success and application volumes
- Delivered interactive visualizations and a project site, earning 1st place out of 4 teams and presenting findings to faculty judges

Moon lander Optimal Control (2025) — Modeled trajectories for fuel-efficient lunar landing

- Formulated the landing problem as a constrained optimal control problem and implemented solvers in Python using convex optimization and ODE modeling
- Evaluated solver performance and fuel trade-offs under different constraints, providing insights into optimal trajectory design
- Produced reproducible codebase and technical report to document design trade-offs

Housing Market Analysis (2025) — Examined affordability and pricing trends across regions

- Conducted exploratory data analysis on national housing datasets, cleaning and transforming data into reproducible notebooks (Python, pandas, matplotlib)
- Analyzed affordability trends and demographic correlations, visualizing drivers of regional market variation
- Presented findings in a structured report with clear visuals to guide interpretation

SKILLS

- **Machine Learning & Statistics**: Python (pandas, NumPy, SciPy, scikit-learn, XGBoost, statsmodels), Bayesian modeling, econometrics, machine learning, Stata
- **Engineering**: Swift/SwiftUI, SQL/PostgreSQL, C++, Java, Git
- **Cloud & Infrastructure**: AWS (Amplify, Bedrock, AgentCore, Lightsail, EC2, ECS/Fargate, Aurora, RDS), Docker
- **AI & Agentic**: LLM applications, MCP servers, multi-agent systems
- **Mathematics & Optimization**: Convex/Dynamic Optimization, Linear Algebra, PDEs/ODEs
- **Language**: Proficient in American Sign Language (ASL)